

Topology optimization of rotor design in switched reluctance motor using immune algorithm

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Abstract. A topology optimization of rotor design in switched reluctance (SR) motors based on immune algorithm and ON/OFF method to obtain practical shapes is presented in this paper. The topology optimization using evolutionary algorithms such as genetic algorithm often obtains complex shapes that are impossible to produce in engineering sense. To solve this difficulty, a filtering process is introduced in the present algorithm based on the immune algorithm. The current waveform in SR motors strongly depend on the inductance that varies by the shape of rotor. This results in high computational cost of numerical analysis to obtain the torque property. In this study, a simple estimation method to calculate the current waveform is introduced. This method enable to reduce the computational cost and perform the topology optimization in a practical time. The present method is applied to a topology optimization of SR motor. Optimization results show that feasible shapes are obtained and torque property is improved.

Keywords: Switched reluctance (SR) motors, on/off method, topology optimization

1. Introduction

Due to the growing awareness of resource and environmental issues, industrial motors are strongly required to be highly efficient. In the field of home appliances in particular, the products what aimed at energy saving, high efficiency, and resource saving have been developed. In addition, motors installed in home appliances account for the majority of power consumption, and technology development for energy saving has been progressing in recent years. The motors determine the basic performance of home appliances such as air conditioner and are required to save energy and improve efficiency. For the above reason, the demand for high efficiency motors using rare earth permanent magnets is increasing. However, using the rare earths has some difficulties: reserves of rare earths are very small, expensive rare earths raise the price of motors and the supply is unstable. As one solution to solve these problems, a switched reluctance (SR) motor that does not use the rare earth magnet has attracted attention [1–3].

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The SR motor is an electric motor that uses reluctance torque caused by changes in the magnetic resistance in the magnetic circuit. Both the stator and rotor have salient pole structures, and the winding is concentrated on the stator only. In addition, since no permanent magnets are used, the structure is simple and robust compared with induction motors and permanent magnet synchronous motors. The applications of SR motor are expected as an inexpensive variable speed motor [4].

However, the rotational force of the SR motor fluctuates at low speed. In addition, a suction force acts in the radial direction, which causes noise and vibration. It has the large fluctuations in torque in low-speed operation [5]. In order to improve the torque property of SR motor, the rotor shape is optimized by using a topology optimization (TO) method.

The TO method can bring novel shapes because this approach searches the optimal shapes by making free modifications in the distribution of material in contrast with parameter optimization [6]. The level set method with sensitivity analysis is a well-known method for TO. However, the optimized shapes obtained by this method sometimes fall into local optima due to poor global search ability [7]. On the other hand, the on/off method is another approach for TO [8]. The authors have been proposed a TO method based on the on/off method with immune algorithm which is a kind of evolutionary algorithms and inspired from theories about the mammalian immune system [9].

In this paper, we apply the present method to a SR motor to improve the torque property: maximizing average torque and minimizing torque ripples.

2. Formulations

2.1. Topology optimization using on/off method

TO is a technique for finding the optimal shape from among the design shapes and has a higher degree of freedom and may yield shapes that are not assumed by the designer.

In this study, an on/off method is used as shape representation method. The on/off method is a method for expressing the material of each element by on/off. In this study, so as to combine the finite element analysis, finite elements are used for the on/off method: ON is treated as a magnetic region and OFF as an air region.

2.2. Immune algorithm

The immune algorithm for TO is inspired from the Clonal Selection Principle, and combines local and global search characteristics in order to avoid falling into local optima. The on/off method uses a 2D map for representing the material distribution in search space. Each finite element is associated to a binary value indicating either the presence or absence of material. The procedures of the present method are summarized below.

- (1) Generate an initial population
- (2) Evaluate the objective function for each antibody.
- (3) Test a stop criterion. If it is satisfied, stop the procedures.
- (4) Eliminate low-ranking antibodies.
- (5) Generate clones for each surviving antibodies.

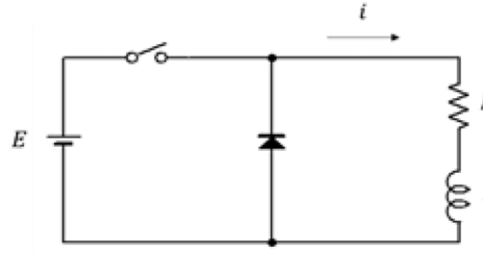


Fig. 1. Equivalent circuit for one phase.

- (6) Small-modifications called affinity maturation on surface are applied to the clones, which are then evaluated over the objective function. Only the best candidate solution from each subset of (parent antibody + clones) is allowed to survive to the next generation.
- (7) Add randomly generated antibodies to replace the ones eliminated in Step 4, in order to keep the population size constant.
- (8) Back to step 2.

2.3. Current calculation method

The basic principle of motor torque generation is based on the force generated by passing a current through a magnetic field. However, the reluctance motor uses only the reluctance torque that is the force generated by the change in the position of the magnetic energy W . When one phase of the stator winding is excited, a magnetic flux φ is generated, and the rotor rotates to the position $\theta = 0$ where the magnetic resistance is minimized. The magnetic entrainment energy W is expressed as follows.

$$W = \frac{1}{2} \varphi i \quad (1)$$

where i is the current in coil. The inductance of the winding with respect to the rotor position θ is defined as $L(\theta)$. The flux is written as $\varphi = L(\theta)i$, and W is expressed as,

$$W = \frac{1}{2} L(\theta) i^2. \quad (2)$$

Since the derivative of the energy displacement is torque, the torque T can be expressed as

$$T = \frac{\partial W}{\partial \theta} = \frac{1}{2} i^2 \frac{dL(\theta)}{d\theta}. \quad (3)$$

Figure 1 shows an equivalent circuit of SR motor for one phase. We can express the voltage equation as

$$E = Ri + L \frac{di}{dt}. \quad (4)$$

The current can be obtained by solving the voltage equation and shown as,

$$i(t) = \begin{cases} \frac{E}{R} (1 - e^{-\frac{t}{\tau}}), & \text{switch on} \\ \frac{E}{R} e^{-\frac{t}{\tau}}, & \text{switch off} \end{cases} \quad (5)$$

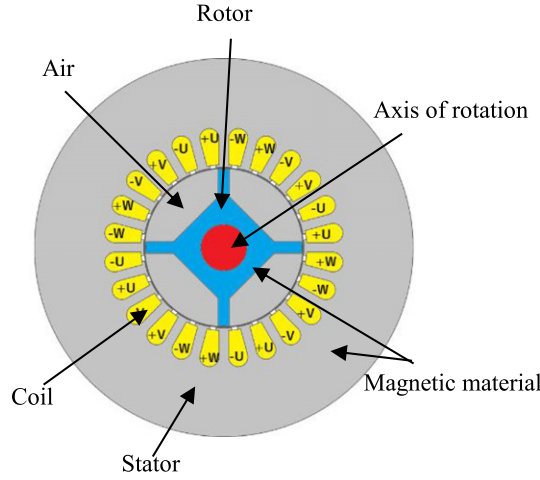


Fig. 2. Comparison model.

where $\tau = \frac{L(\theta)}{R}$. The inductance $L(\theta)$ depends on the shape of rotor that is target of TO and is changed during the optimization process. Therefore, a combined analysis: computation of $L(\theta)$ by magnetic field analysis and electric circuit analysis should be performed. This results in high computational cost. To solve this difficulty, a simple estimation method is introduced. The finite element analyses are performed in each mechanical angle $\theta^i (i = 1, 2, \dots, n)$. The current $i(\theta)$ is estimated from $\tau(\theta^{i-1})$, and then the $L(\theta^i)$ is calculated by the finite element analysis with $i(\theta)$, and $\tau(\theta^i)$ is obtained for the next mechanical angle $i + 1$.

2.4. Current calculation method

The model of the SR motor handled in this study is shown in Fig. 2 and it is based on the Institute of Electrical Engineers of Japan (IEEJ) D model [10]. The differences from the original D model are that the permanent magnet is taken out and treated as an air region, and the fixed shaft is closed with a magnetic material. The major specification of this motor is that: a stator diameter of 112 [mm], a rotor diameter of 55 [mm], a 4 poles machine, and a 3 phases, 24 slot coils. In addition, since a magnetic flux distribution periodicity appears every 1/4 period, the analysis is performed only in the 1/4 region. This model is used for a comparison model. In Fig. 3, the shape obtained in the design area is copied and analyzed so that it is line symmetric with respect to the shape symmetry axis.

3. Optimization results

We performed an optimization of SR motor to maximize the average torque and minimize the torque ripple. The objective function F is defined by,

$$F = k \frac{T_{ave}}{T_{ave0}} (1 - k) \frac{T_{rip}}{T_{rip0}} \quad (6)$$

where k is a weight coefficient, T_{av} and T_{rip} are the average and ripple of torque. T_{ave0} and T_{rip0} are the average torque and the torque ripple in the comparison model. A typical current waveform is shown in

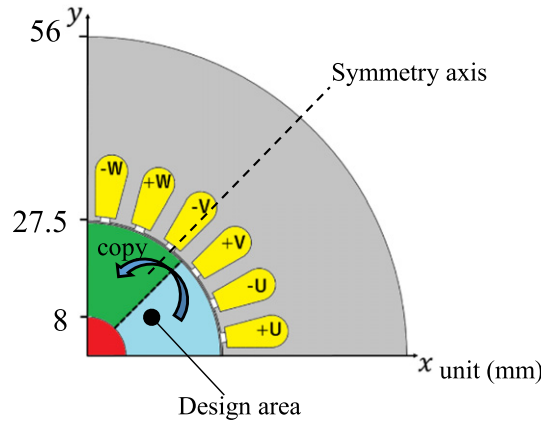


Fig. 3. Model of SR motor for optimization.

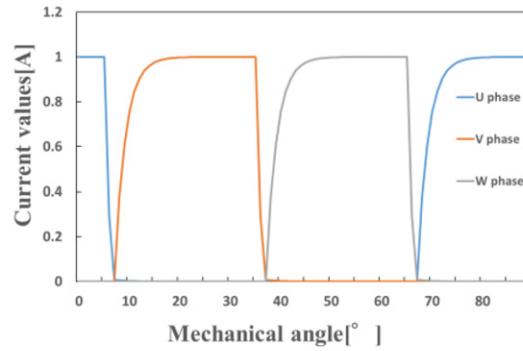
Fig. 4. Example of current waveform in the comparison model with $E/R = 1$.

Fig. 4 and the current is set to U , V , and W three phases. First, the current is supplied to the U phase, the rotor is rotated counterclockwise, and the current is sequentially switched from $U \rightarrow V \rightarrow W$ every 30° mechanical angle. In addition, the current is set as exponential with relaxation time that is calculated by finite element analysis at the previous time step as described the above chapter. The magnetic material in rotor and stator are modeled by JIS 50A350 (JIS: Japanese Industrial Standard). The TO is performed over 200 generations. The initial shapes are generated by using pseudo random number. Figure 5 shows the graph of objective function and average torque values. It has improved rapidly in about 30th generation, and there is little change after that. Moreover, it has converged after the 150th generation. It seems that the 200 generations are enough to optimize in this model. Both Fig. 5(a) and Fig. 5(b) shows that the present method has good convergence. The optimized shapes in case of $k = 0.3$ is shown in Fig. 6. The optimal shape obtained by the present TO method seems to be feasible. T_{rip} in the optimized model is larger than that in the comparison model. T_{rip} is defined by $T_{rip} = T_{max} - T_{min}$ where T_{max} and T_{min} are the maximum and minimum of torque, respectively. T_{max} as well as T_{avg} are increased by the optimization. However, T_{min} cannot be improved and $T_{min} = 0$ in both the optimal solution and the comparison model. Therefore, the T_{rip} in the optimized case is larger than that in the comparison model. The T_{min} can be improved by changing of on/off timing. We have performed optimizations with different value of k . In case of $k = 0.1$ or 0.2 , a meaningless optimal solution with $T_{ave} = 0$ is obtained.

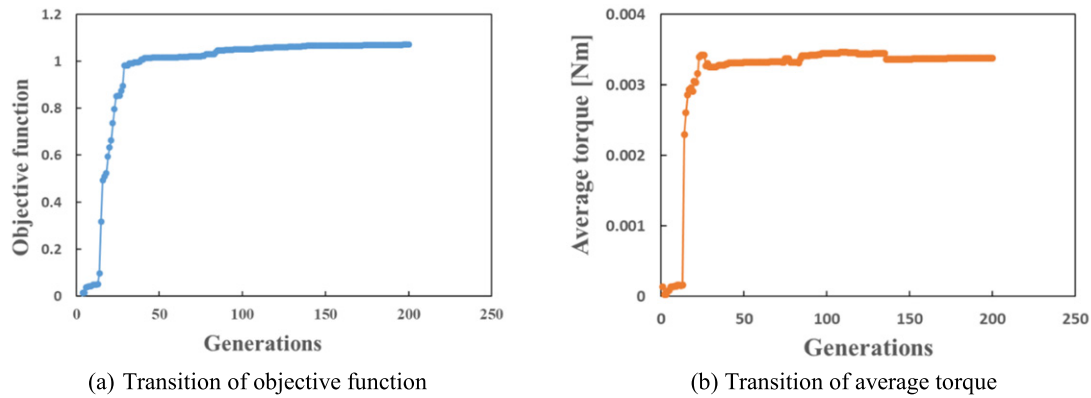


Fig. 5. Transition of objective function and average torque.

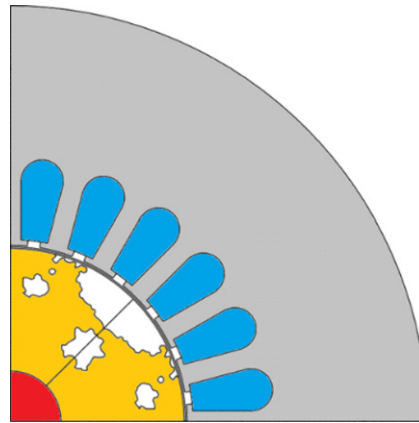
Fig. 6. Optimization results of reluctance motor ($k = 0.3$).

Table 1
Settings of immune algorithm and computational time

Number of population N	5
Generation	200
Total computational time	8 h 30 m

Intel Xeon E5-2650V2, Intel C++ compiler, Linux OS.

The computational time and settings of TO are summarized in Table 1. We can see that the optimized shape can be obtained within one day. Figure 7 shows the magnetic flux line distribution between the comparison model and optimized model when the mechanical angle is 8° . We can see that the rotor of the optimized model has a large pulling force from U phase to V phase. This is the reason for the increase in torque. Figure 8 shows the optimization result of the torque curve in mechanical angle, and the values of torque are shown in Table 2. We can see that the torque ripple increased slightly, but the average torque increased by 462%.

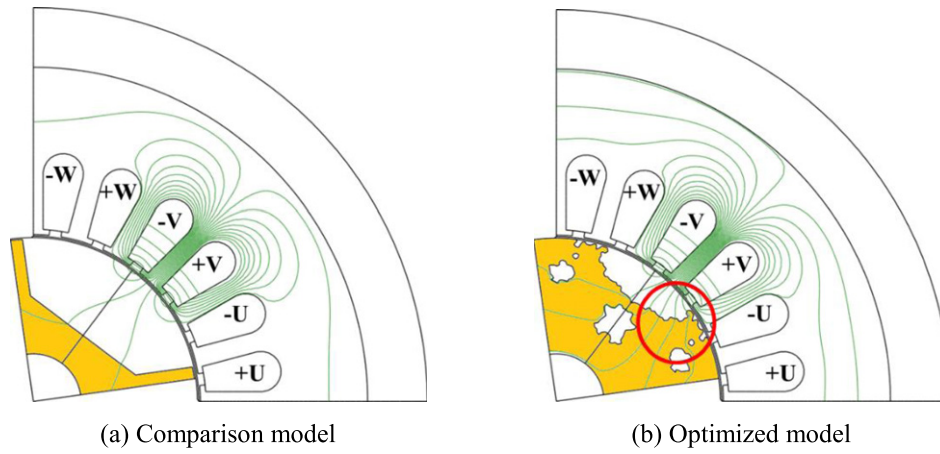


Fig. 7. Magnetic flux line distribution comparison (Mechanical angle of 8°).

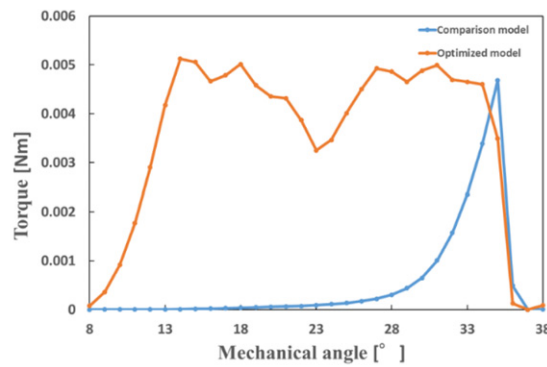


Fig. 8. Torque curve in mechanical angle.

Table 2
Optimization result of torque comparison

	Comparison model	Optimized model	Proportion (%)
T_{ave} (Nm)	7.311×10^{-4}	3.777×10^{-3}	461.9
T_{rip} (Nm)	4.687×10^{-3}	5.126×10^{-3}	109.4

4. Conclusion

The purpose of this study is to improve the average torque and reduce the torque ripple of switched reluctance motors. In this study, a topology optimization method by using immune algorithm is introduced so as to obtain feasible optimized design. The optimization successfully results in improving the average torque. In future, the optimization of switching timing as well as rotor shape will be performed.

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